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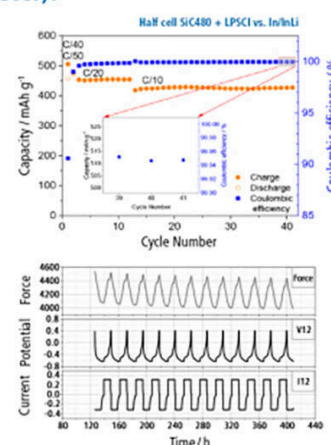
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Machine vision for detection of defective oil palm fresh fruit bunches using YOLO Algorithm

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Abstract. In the last couple of years, machine vision with object detection methods has been applied in agriculture, especially using deep learning with the You Only Look Once (YOLO) algorithm. It has gained attention for rapid in-line sorting machine vision, where efficiency in time and labor cost is crucial. This study designed and built a real-time machine vision system and a YOLOv4 tiny algorithm to identify defective oil palm fresh fruit bunches (FFBs) moving in a roller conveyor. Oil palm FFBs are the source of crude palm oil (CPO), which is the export commodity for countries of Indonesia and Malaysia. Sorting FFB is a daily process when an oil palm FFB transporting truck arrives at an oil palm refinery mill. This process is usually done manually and traditionally by two experienced workers who sort for unqualified and defective FFBs, such as long stalk, rotten, empty, and thorny or spiky FFBs. We developed a two-class model to detect normal and defective FFBs. The model has obtained the average precision (AP) of 98% for normal and 96% for defective oil palm FFBs. The accuracy validation using the confusion matrix resulted in an mAP of 99.7%. The results exhibited the potential of machine vision and YOLOv4 tiny to detect defective FFBs.

Keywords: Defect Detection, Machine Vision, Oil Palm Fresh Bunches, Two Class Model, YOLOv4



1. Introduction

Computer vision and automation are two related terms in agriculture, where efficiency in time, labor, and operational costs is crucial. The terms are also related to precision agriculture and smart farming. Precision Agriculture has developed rapidly as an essential field nowadays, since the swift advancement in computer vision methods, image processing techniques, and camera types. Precision agriculture aims to increase yield, labor, and operational cost efficiency, and reduce environmental impact. The priority aspects of precision agriculture are automatic early detection of plant diseases, robotic fruit and vegetable harvesters, post-harvest sorting, and grading. Precision agriculture is related to smart farming, which uses computer vision, information and communication technology, and artificial intelligence (AI). Computer vision uses a computer and a camera to simulate human vision. Image processing and machine learning mimic the human brain in making decisions. Information and communication technology enables farmers to monitor and evaluate crops remotely, hence easing field and production management. Computer vision helps to overcome many problems in industrial agriculture [1,2,3].

Computer vision (CV) and machine vision (MV) have applications in post-harvest sorting and grading of fruits and vegetables. CV and MV are different scopes. Machine vision is often used in manufacturing or in-line sorting and grading processes of agricultural products. It consists of a computer vision camera and algorithms, equipped with a lighting unit, moving conveyors, control units, and separators. Computer vision has a broader scope, not only involving cameras such as in visible, hyperspectral, multispectral, and fluorescence imaging but also other sensor types such as thermal and Lidar imaging. Post-harvest sorting and grading machine vision aims to substitute the traditional and manual practices by experienced graders, which are time-consuming, labor-intensive, and prone to subjectivity and error [4]. A machine vision system is a rapid, non-destructive, cost-effective, and reliable method for inspecting fruit and vegetables. The inspection includes in-line sorting based on standard classification, detecting contaminants and defects, and estimating chemical contents. The types of inspections rely on the computer vision methods used, which range from colour imaging to spectral imaging, from UV to IR wavelength regions [5].

Machine vision with machine learning has become a suitable method for rapid, cost-effective, and non-destructive detection or inspection of fruit defects in the in-line sorting and grading process. Fruit defects, especially skin effects, affect the fruit's price and consumer preference. The skin defect is more difficult to detect than other external quality attributes such as colour, size, and shape. Challenges still exist for available real-time defect detection technologies, especially in reducing processing time and integrating many image types [6]. There are many types of imaging methods used to acquire and process the defective fruit images, including color imaging and hyperspectral imaging. Machine vision with color imaging is a simple and economical technique for detecting fruit defects. The color imaging has been able to detect the calyx and stalk scar of tomatoes by an RBF-SVM classifier [7]. Hyperspectral imaging has also enabled blueberry internal bruising detection using hyperspectral transmittance images and fully convolutional networks (FCN) [8]. Another technique of machine vision was to use a depth camera to predict dimensions, mass, and defects of potatoes [9].

Machine vision with object detection methods has attracted more attention for defect detection, due to advances in deep learning (DL) techniques. The development has facilitated many machine vision applications in the sorting and grading of fruits and vegetables. Machine

vision with color imaging and object detection techniques is a simple, real-time, and low-cost system for defect detection. This combination method is based on video frames. Several DL algorithms have been developed for these purposes, including a series of convolutional neural networks (CNNs). There are some CNN models available for fruit image processing, such as AlexNet, GoogLeNet, LeNet, and ResNet [10]. ResNet, DenseNet, MobileNetV2, NASNet, and EfficientNet were tested to find the best model for the grading of apples and bananas [11]. Region-based CNN (R-CNN) is one of the CNN series that is related to object detection. Key performance parameters of real-time object detection techniques are frame per second (FPS) and Mean Average Precision (mAP). Developments of CNN algorithms towards higher FPS and mAP led to Fast R-CNN and Faster R-CNN series. However, the You Only Look Once (YOLO) algorithm is the best of the CNN family, which has revolutionized object detection techniques [12].

The YOLO algorithm has become one of the accepted object detection algorithms. It is due to its simplicity and low cost if used with color imaging in video format. The early applications of the YOLO algorithm are fruit detection, tracking, and counting using video frames. YOLO based apple yield estimation was one of the applications that aimed to substitute manual yield estimation. The estimation has succeeded in detecting and counting the apple fruits in vertical fruiting-wall architecture using a detection model of apples and tree trunks with a YOLOv4-tiny network [13]. Another YOLO application was using two-stage image processing for fruit counting based on video frames [14]. The YOLO algorithm types used for ripeness level detection, which use four convolutional neural networks (YOLOv3, YOLOv3-Tiny, YOLOv3-SPP, and YOLOv4). The algorithm models were trained in the Darknet deep learning framework and compared [15].

Palm oil industries are the key export commodities for some countries, such as Indonesia and Malaysia. The quality of crude palm oil (CPO) depends on the quality of its raw materials, which are oil palm fresh fruit bunches (FFBs). The sustainability of these industries needs optimistic efforts. The Industry 4.0 concept is one of the efforts that promotes Artificial Intelligence (AI), Internet of Things (IoT), and cybersecurity. Automatic processes in many stages of palm oil mills are crucial to ensure the palm oil quality [16]. The sorting and grading process is essential for oil palm FFBs that arrive at the receiving stations of palm oil mills. This activity is done manually and traditionally by experienced personnel. The sorting and grading process starts as an FFB transporting truck entering the mill sorting or receiving area. It usually takes four workers to work for one FFB truck. Two workers unload the oil palm FFBs, and the other two workers sort the FFBs according to standard guidelines. Some of the regulations include four ripeness stages (unripe, underripe, ripe, overripe), damaged (by animals or during harvesting or transporting), long stalk, and small (less than 3 kg) [17]. Other criteria are thorny, rotten. Thorny FFBs show more spikelets than fruit (poorly pollinated), while rotten ones show a dirty and wet bunch. The workers use axes and crowbars to pull and move the FFBs aside. These practices are tedious, laborious, subjective, and prone to error and dispute. Therefore, machine vision with computer vision and deep learning offers an easy way to substitute the traditional and manual sorting process.

There have been many efforts to use non-destructive methods for oil palm FFB quality evaluation. Most quality evaluations of oil palm FFBs have used maturity or ripeness levels as the quality attributes. Thermal imaging is one of the optical methods to evaluate the maturity levels of oil palm FFBs [18]. Another non-destructive method was using a smartphone application to calculate harvested oil palm FFBs before they were transported to palm oil mills.

It aimed to reduce the embezzlement of oil palm FFBs during transportation [19]. A combination of computer vision and laser-light backscattering imaging was able to evaluate the classification of oil palm FFB maturity levels [20]. Spectral imaging, such as multispectral imaging, has also found applications in oil palm FFB ripeness classification [21].

Computer vision with object detection algorithms has received significant attention in the last couple of years in the oil palm industry. With the development of the YOLO algorithm series, more applications have been proposed for oil palm FFB classification and detection. Many automatic detection systems have used the YOLO algorithm for oil palm FFBs, such as an optimized YOLOv3 tiny to detect and localize three objects at a palm oil plantation under various environmental conditions [22]. Another application was with the YOLOv3 algorithm to detect ripeness levels of oil palm FFBs in real time [23]. The next application was using several object detection algorithms for FFB ripeness classification based on fruit surface colors [24]. Another application was counting of oil palm FFBs on a moving conveyor using a three-step algorithm: detection, tracking, and counting [25]. Computer vision and YOLOv4 were the first steps toward a real-time robotic FFB harvesting system [26]. A comprehensive annotation and labelling process was performed to build a dataset of 6 maturity categories of oil palm FFBs [27].

This study aimed to build a real-time machine vision system to simulate the sorting process at the receiving station of palm oil mills. This study makes a new contribution to YOLO-based computer vision applications for evaluating oil palm fresh fruit bunches (FFBs), particularly in detecting and sorting defective FFBs on a moving conveyor. Most applications of the YOLO algorithm were for the FFB harvesting process. Most studies evaluated FFBs based on ripeness levels. FFBs received at the palm oil mill area originated from many surrounding plantations. The FFB conditions are varied due to long road distances from the plantation to mills, rain, and damaged roads that cause damage or defective FFBs when they arrive at mills [17]. In this study, the defect categories were based on palm oil mill regulations, which sort four defect categories: long stalk, thorny, empty, and rotten bunches. The FFBs from other categories have been sorted out before arriving at the receiving stations. The machine vision system consisted of a computer vision unit and a 5-meter roller conveyor. The conveyor was equipped with a motor, control unit, computer, Arduino microcontroller, user interface, and a separator. We utilized a CMOS camera that is integrable into the machine system hardware and software; hence, the sorting process is fast and real-time. A confusion matrix was used as the metric to evaluate the model's performance.

2. Material and Methods

2.1. Machine Vision System

Detecting defective oil palm FFBs is one of the machine vision tasks that aims to simulate and substitute the manual sorting process at the receiving area of palm oil mills. This study has developed machine vision hardware and software for detecting defects of oil palm FFBs according to the government regulation on FFB buying criteria for small holders as well as the private companies, which includes rotten, empty, long stalk, and thorny or undeveloped. Other criteria, such as muddy, wet, damaged spikelets, stacked or partially occluded FFBs, were not included in this study, as they were found in fewer numbers in the sorting areas. The system uses computer vision and object detection methods with the YOLO algorithm. The machine vision system and its components are shown in Figure 1.

The system in Figure 1 consists of a color camera and lens, a light source, a conveyor unit, a pneumatic-based arm or separator, a control unit for the motor, and a separator. The system also has an Arduino board connected to computer units and actuators. The actuator is connected to the control unit to initiate the conveyor and the separator lever. The computer vision unit had white metal shielding that limited the background light entering the camera. The shielding has frames to position the camera, lens, and lamps. The lighting came from a pair of neon lamps with 5 W of light power. The computer vision unit has some essential components, including the camera and lenses. We used DFM 22BUC03-ML USB 2.0 color board camera, equipped with a 16 mm 1/3" M-12 lens. Some requirements for a real-time camera are a USB connection, a trigger, I/O inputs, and a compatible camera driver. The camera driver should be able to be embedded in a written acquisition program. The CMOS color camera has a resolution of 0.4 MP (744 x 480 pixels) and a frame rate of 76 FPS. The camera and lens were positioned at the middle computer vision unit with a 90 cm distance from the lens to the conveyor surface (about 60 cm working distance). The conveyor has a white plate floor to increase background segmentation efficiency in image processing. Some software used includes Python 3.8, YOLO v4 (tiny), SORT for object tracking library, OpenCV for Computer Vision Library, and Arduino IDE for Microcontroller programming. We also used Flask for a Web Server Library for web-based/remote detection and monitoring. We used two separate computers: the main computer (Intel NUC), which connected to the camera and other sensors, and another computer for real-time Arduino-based control of the pneumatic control valve.

The machine vision for defect detection runs in real-time. The logical commands for the defective FFB detection model start as follows. When the camera detects an FFB on a video

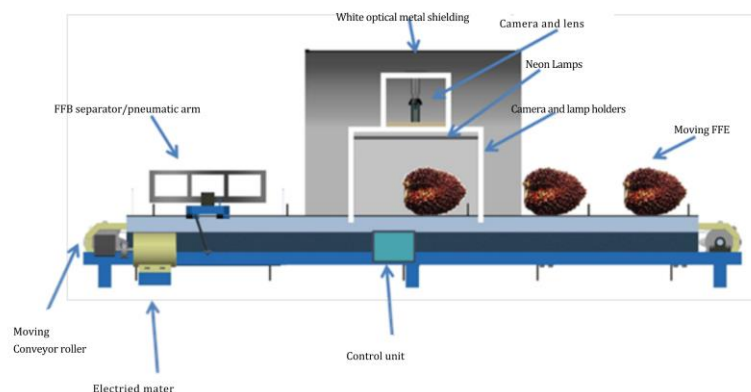


Figure 1. Machine vision system for defective oil palm FFB

frame, a bounding box is assigned and then tracked. Tracking is in progress until it reaches the kill line of kill & trigger. If the object passes the limit line, the kill & trigger protocol is performed, which has an abnormal class > 70%. Then the signal "1" is sent to the Arduino input, which waits for 6 seconds. Until pin "8" at Arduino receives "digitalWrite 1". When pin 8 receives the signal, the pneumatic valve that controls the separator will move forward, aiming to push the defective FFB out of the conveyor line. After 500 ms, Arduino will send another signal to the pneumatic valve to withdraw. If the object bounding box passes through the limit line of kill & trigger is a normal object, the Arduino does not receive any signal.

2.2. YOLO model construction

There are some stages to obtain a YOLO detection model for defective oil palm FFBs. The first stage was setting up and characterizing the computer vision unit, as in Figure 1. The second stage was to build the detection program. A defective FFB detection program was developed using Python language and OpenCV Library. The third stage was to construct a YOLO model. We used the YOLO algorithm due to sorting efficiency. In a reception area, it takes about 30-45 minutes to unload and sort oil palm FFBs from a transporting truck. The fourth stage was to integrate mechatronic circuits and control into the acquisition and YOLO detection program.

Figure 2 shows the steps for constructing the two-class defective detection model. First, FFB images were acquired to build a dataset [28], which used oil palm FFBs from a PTPN V plantation of Riau Province, Indonesia, by bringing the machine vision system to its receiving area. Three experienced graders at the mill validated the FFB classes before operators recorded FFB images. The images were saved in two separate file folders for the annotation stage. The annotation process used the Labellmg software, resulting in files with the extension.txt. Next, all the annotated images were divided into training and testing datasets with a ratio of 80:20 %. The training dataset was uploaded to Google Drive for YOLOv4 tiny Model training using the Cloud GPU platform. Later, the trained model information is inserted into the defect detection program to identify the defective oil pal FFBs. In the testing stage, the model applied the 20% dataset. Finally, the model's performance was evaluated using the confusion matrix, and the YOLO configuration was updated. The filter parameter size used for YOLO configuration during the training process was $(5 + n_class) * 3 = 21$.

2.2.1. Dataset Preparation

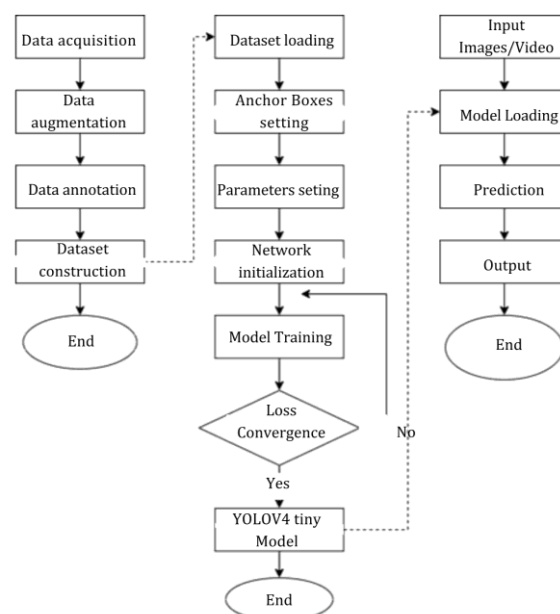


Figure 2. Flowchart for YOLO model construction

Dataset preparation is the crucial part in building a YOLO detection model. The dataset images were obtained using validated FFBs at a government-owned palm oil plantation, PTPN V, Riau Province, Indonesia. The detection model classified the oil palm FFB samples as normal and defective FFBs. In this study, ripeness level features are excluded. The defective FFBs consist of four categories according to standard criteria, as rotten, long stalk, empty, and thorny or undeveloped bunches [17], as shown in the Figure. 3. Figure 3 shows the FFB images taken using the IMS camera (744 x 480 pixels). We adjusted the raw image using the correction settings for each category detail (no pre-adjustment was made for the dataset images). Figure 3 displays four categories of defective FFBs and the variation of each defect category. Each class of normal and defective had 200 oil palm FFBs; each defect category had 50 FFBs.

Image acquisitions were performed on the conveyor using the computer vision unit. Each

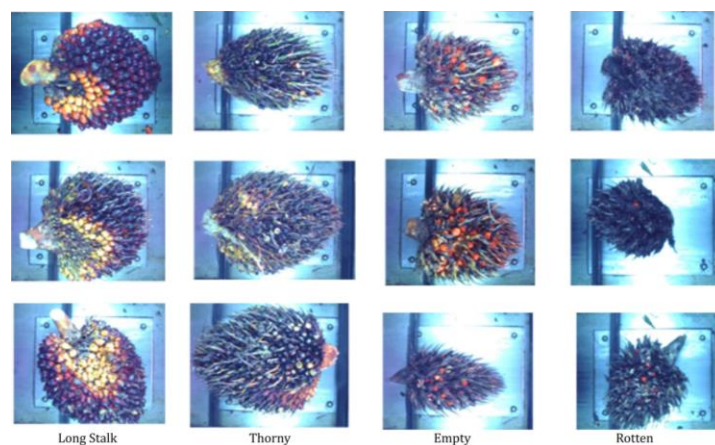


Figure 3. Defect categories and variation (long stalk, thorny, empty, and rotten) oil palm bunches

FFB had 6-8 images, which consisted of a back and front bunch. The front bunch is the part facing sunlight when intact on a palm oil tree, having higher fruit density, while the back part contains more infertile fruits and is lighter in color. For each part, the images were taken at four positions within the camera's field of view (FOV). Therefore, each bunch had 6-8 images. These different positions and parts aim to anticipate different orientations and positions of FFBs on the moving conveyor. Therefore, the two-class model had a dataset of 2400 images. We divided the dataset into a ratio of 80:20% for training and testing. Figure 3 shows the physical characteristics of the defective oil palm FFBs. Long stalk FFBs are for unripe, ripe, and overripe levels that have stalks longer than 2 cm. Thorny FFBs are undeveloped FFBs having long spikelets and small fruit size. An empty bunch means fewer fruits left in the bunch, while rotten or damaged bunches have darker brown spikelets and fruits [17, 27].

The annotation step is essential in building a detection model. Annotation is a labeling process using a labeling tool that aims to prepare the detection program to identify objects in a set of video frames. The labeling step creates an object bounding box and class name in an image, as displayed in Figure 4 (a). One of the available labeling tools is the Python application program Labellmg. The Python software obtains a file in .txt format, which contains image data. Each bounding box has a size that is different for each object. The bounding box size is

represented by its coordinates, shown at the bottom of Figure 4. Figure 4 (b) shows the annotation results for two classes of FFBs. The text file contains the bounding box information, which includes classIndex, xcen, ycen, w, and h. The first information is the class index or object class, in this case, normal and defective. The other four numbers indicate the position and size of the bounding box (six digits and below zero) with w and h of width and height, respectively [27, 28].

2.2.2. Detection model

The detection model for identifying normal and defective oil palm FFBs used the YOLOv4tiny algorithm. YOLOv4 algorithms have applications in fruit and vegetable sorting machine vision due to their more accurate and shorter processing time compared to other object detection algorithms. YOLOv4 tiny was the improvement of YOLOv4 to increase speed and decrease computational cost [29]. The parameters of the YOLOv4 tiny configuration file for oil palm FFB detection model include batch number (64), subdivisions (16), width (416), height (416), classes (2), and filters (21). The parameters we used are generic from the Darknet framework. Some parameters, such as subdivision (16), max_batches (6000), classes, and filters, were changed according to the needs of the defect detection models. Max_batches = 6000 was due to the number of classes being less than 3 for the two-class model. The steps consisted of 80 % and 90% of Max_batches.

Training and testing are the final two steps for constructing an object detection model.

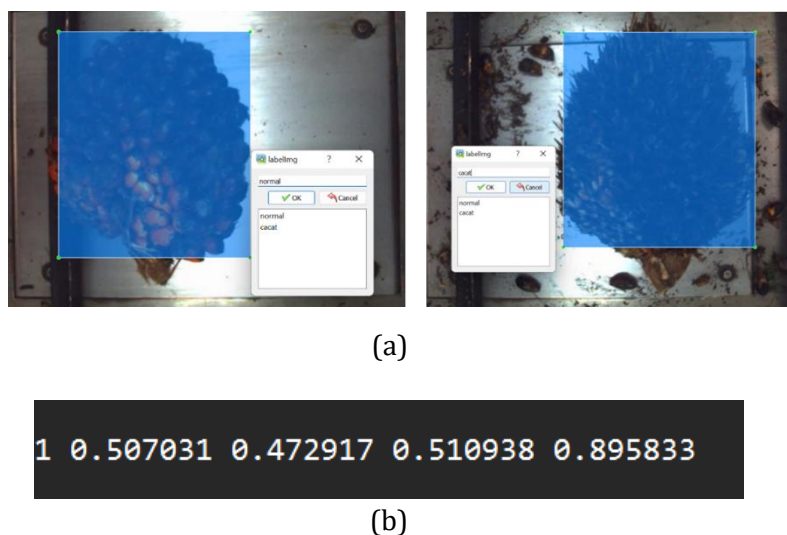


Figure 4. Annotation of FFB using Labellmg application (a), Image annotation results as file.txt(b)

These processes used Darknet and Google Colab. Darknet has a data file, a configuration file, and pre-trained weights. The data file is the custom dataset of annotated images of normal and defective oil palm fresh fruit bunches (FFBs). The dataset was split into two training and testing datasets with a ratio of 80:20 %. The two-class model used a dataset of 2400 images. The YOLOv4 tiny model was pre-trained and fine-tuned using the configuration file as shown in Table 1 via the Darknet platform. Later, the optimal model with pre-trained weights was trained and tested using Google Colab, and its performance was measured.

2.3. Performance analysis

There are three main parameters used for model performance analysis, i.e., average precision (AP), mean average precision (mAP), and intersection over union (IoU). AP and mAP values are parameters of the confusion matrix, often used for any deep learning methods. Confusion matrix for the object detection model relies on detection results. IoU is another metric for object detection models, which tells how overlapping the predicted and the ground truth bounding boxes of an object are. IoU is measured as the ratio of the overlapping bounding box area and the combined bounding box area, as written by Eq. (1). IoU has values of 0 – 1 or %, which relates to 0 as no intersection and 1 (one) as intersecting perfectly [28].

$$IoU = \frac{\text{area of overlap}}{\text{area of union}} \quad (1)$$

Confusion matrix is a value matrix that compares predicted and actual values (ground truth) using four components: true positive (TP), false positive (FP), true negative (TN), and false negative (FN). For the two-class model, positive and negative conditions refer to normal and defective FFB states. TP means a normal sample from the testing dataset is classified or detected as normal correctly. In the same manner, TN refers to the defective sample being correctly detected as defective. FP means a defective sample is wrongly detected as normal, while FN refers to a normal sample falsely detected as defective. False Negative (FN) refers to a sample belonging to the positive class but being classified wrongly as belonging to the negative class. The parameters of AP, mAP, and IoU are related to the four components. The confusion matrix parameters of a YOLO detection model can be computed using the scikit-learn package of Python.

Other metrics defined using TP, FN, and FP are Accuracy, Recall, Precision, and F1-Score. Recall aims to measure how many actual positives (ground truth) are predicted correctly by the detection model using Eq. (2). In contrast, Precision represents how many predicted positives are correct compared to the total tested samples. Precision is calculated using Eq. (3). F-score is intended to measure the balance between Precision and Recall as presented in Eq. (4).

$$\text{Recall (R)} = \frac{TP}{TP + FN} \quad (2)$$

$$\text{Precision (P)} = \frac{TP}{TP + FP} \quad (3)$$

$$\text{F1-score} = \frac{2 \times (\text{Precision} \times \text{Recall})}{\text{Precision} + \text{Recall}} \quad (4)$$

Average Precision (AP) and Mean Average Precision (mAP) are two parameters often used to measure a detection models. Both parameters can be calculated from Recall and Precision values as written respectively in Eq. (5) and (6), in which n is the total number of objects or testing samples [29].

$$AP = \sum (\text{Recall}(n+1) - \text{Recall}(n)) \times \text{Precision} \times \text{Recall}(n+1) \quad (5)$$

$$mAP = \frac{1}{n} \sum_{k=1}^{k=n} AP(k) \quad (6)$$

3. Result and Discussion

This study aimed to design YOLOv4 tiny models for the computer vision unit to detect normal and defective oil palm FFBs on a moving conveyor. The unit is a part of a machine vision system for sorting and grading oil palm FFBs. FFB sorting process looks for FFBs that meet standard criteria besides ripeness levels, such as long stalk, empty, thorny, and rotten FFBs, namely as defective FFBs. The two-class models divided the FFBs into normal and defective.

3.1. Training results

Figure 5 shows the graph of Loss and mAP versus iteration number. It depicts that the model has converged, where the Loss value reaches a minimum at 5900 (0.04), with a score of 99.71%. The Loss converges almost to 0, meaning the model has learned during the training phase. However, it cannot be translated as the model accuracy. The model accuracy will be evaluated on the test set. The input of this process was from an 80% dataset. The output of this process was weight files, which will be used for the testing and validation stages. In this stage, transfer learning was performed, aiming to use the trained model to solve the same case in other instances. Transfer learning for this stage was obtained from the AlexeAB repository, which was trained before on the Coco Dataset, the dataset from Microsoft that has 80 classes. We used YOLOv4-tiny of yolov4-tiny.conv.29

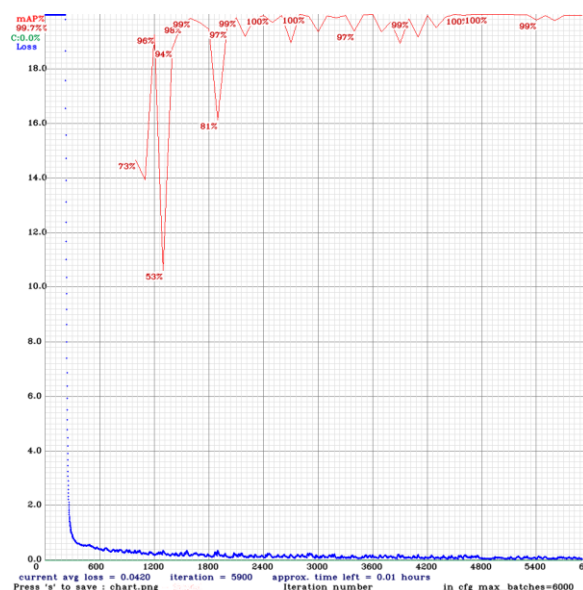


Figure 5. Graph of Loss and mean Average Precision (mAP) versus iteration

Table 1 shows the model performance of YOLOv4 tiny version in 1000 iteration. The best performance was at 5000 iteration, where the detection model can reach mAP of 99.08% and

Table 1. Training Performance of YOLOv4 tiny two class-model for defective oil palm FFBs

Model Iteration	mAP(%)	Average IoU(%)
3000	96.53	78.44
4000	98.05	80.90
5000	99.08	87.79

average IoU of 87.79%. The results will be used for the testing and validation stages because they obtained the best mAP and IoU scores.

3.2. Testing results

Figure 6 shows the results of the testing stage for normal and defective oil palm FFBs in real-time, using the Python program. The testing stage used the weights file that had the best results from the training stage. Figure 6 shows that the detection model displays bounding boxes for

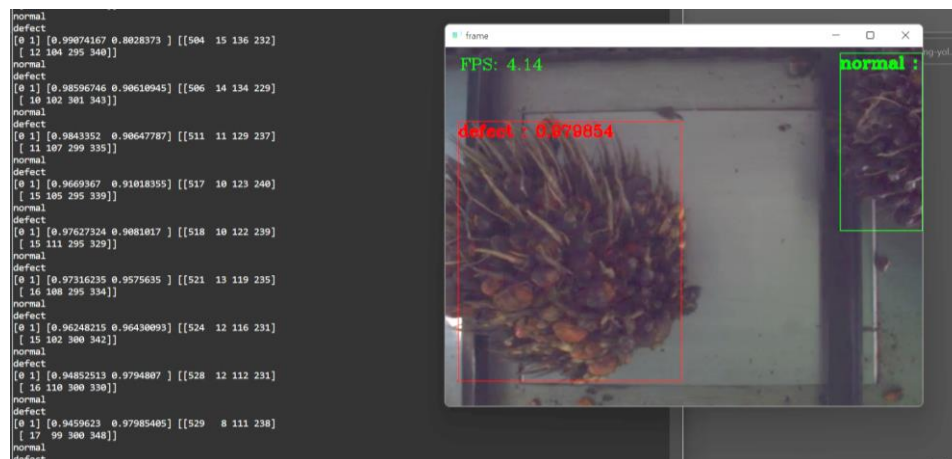


Figure 6. The testing output of detection results using Python Scripts.

the normal class (green box) and defective class (red box). It also shows the accuracy for each bounding box and the speed in frames per second (FPS).

Model validation is the final step for constructing a detection model after the testing stage, as in Figure 2. This stage used video frames of normal and defective FFBs moving on the conveyor, and also performed at the palm oil mill with experienced graders standing by [30]. Figure 7(a) shows that the YOLOv4 tiny model succeeded in detecting the normal and defective oil palm FFBs. The two-class model has detected the oil palm FFBs passing the camera with an average precision of 99.65% for normal and 99.38% for defective FFBs.

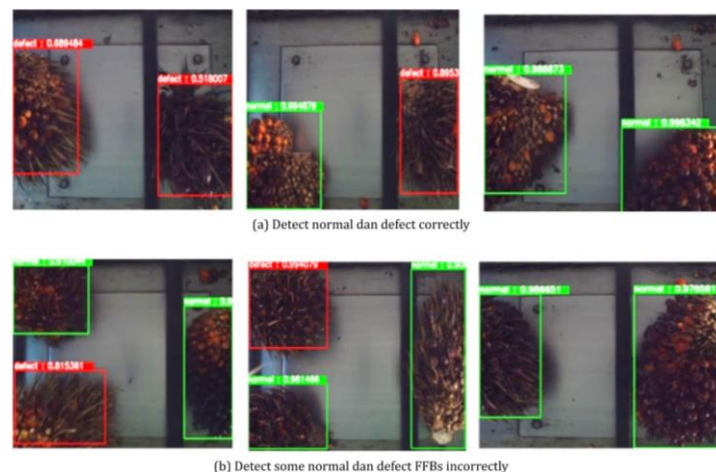


Figure 7. Frames of detection results using YOLOv4 tiny model (a) detected true for normal and defective FFBs (b) Some FFBs detected false.

Figure 7 (b) displays that the model detected some FFBs incorrectly due to several conditions. One reason is that the model experiences overfitting, in which the model performs well during training, but poorly during testing and validation stages. The causes of the problem could be the need for larger datasets, especially for the defective oil palm FFBs. During the study, we waited for many transporting trucks to arrive to have more defective samples [13]. Another cause of overfitting is the lighting unit having lower intensity, making the model unable to differentiate between two classes, also due to different sizes of FFBs, which cause out-of-focus. This problem can be overcome using an auto focus lens [31].

3.3. Confusion matrix results

Confusion matrix was used to measure the model's performance during the validation stage. If a normal oil palm FFB is detected as normal, the condition will be defined as a True Positive (TP) because the detection result is identical to the labeled FFB. If the normal FFB is detected as defective, the condition is defined as False Positive (FP). The same rules are also valid for the labeled defective FFBs. True Negative (TN) is for labeled defective FFBs that are detected defective, otherwise as False Negative (FN).

Table 2 shows the results for the validation stages of the YOLOv4tiny model for 60 FFB samples, comprising 30 FFBs for each class. It shows that the normal class has 90 % precision, close to 100 % Recall, and 94% F1-score. The defective class has 80 % precision, 100 % Recall, and 88% of F1-score. Table 2 also shows that there are some FP and FN scores. The problem can be overcome by using more classes for the detection model. The results show that the YOLOv4 tiny model can detect both classes, but the defective class has less performance. The causes could be due to the need for more images for the four types of the defective class [14, 27].

Table 2. Confusion Matrix Results

Image class	actual	prediction	TP/TN	FP/FN	Precision	Recall	F1-Score
Normal	30	27	27	3	0.9	1	0.94
Defect	30	24	24	6	0.8	1	0.88

4. Conclusion

This study has successfully constructed a machine vision using the YOLOv4tiny algorithm to detect normal and defective oil palm fresh fruit bunches (FFBs). The defective FFBs consisted of long stalks, empty, thorny, and rotten FFBs. The contribution of this study is the real-time machine vision for FFB sorting and grading at the reception stations of palm oil mills. The hardware consisted of conveyor and control circuit units for the conveyor and for the separator arm. The software parts are the detection models, the GUI, and the information system. The training stage obtained a model accuracy of 99.71%. From the testing stage, the detection model has had an mAP of 99.08% and an average IoU of 87.79% at the iteration of 5000. The validation stage obtained a mean Average Precision (mAP) of 99.6% for normal FFBs and 99.7% for defective FFBs.

This study has limitations in that only a class model was considered. The aim was to accommodate one separator arm, so that the normal FFBs will pass, while the defective FFBs will be ejected. The system and the model can be extended into a four-class model, which classifies FFBs as long stalk, empty, thorny, and rotten FFBs. Other than the four classes, the FFBs are considered normal or have passed the standard criteria. Another exploration was to construct a model for stacked FFBs, where several types of FFBs were in front of the camera; the model can simultaneously identify ripe, unripe, long stalk, empty, thorny, and rotten FFBs. However, for these models, other types of separator arms should be considered.

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Author contributions

All the authors have contributed to this study. Dodi Sofyan Arief and M Dalil have contributed to the design and construction of the system hardware, especially the conveyor and the control circuit unit. Minarni Shiddiq contributed to the imaging systems and characterization, obtained the research grant, wrote the manuscript, and is the corresponding author for this manuscript. Jahrizal contributed to the field management when taking data in PTPN V and preparing the Grant proposal. Roni Salambue and Ali Yuhendra contributed largely to building the GUI and information system for the machine vision. Vicky Vernando Dasta and Muhammad Ikhsan Hamid have designed and built the acquisition, image processing, and YOLO model scripts in the Python language platform. Dewi Anjarwati Mahmudah and Ihsan Okta Harmailil were the operators of the machine vision system for training, testing, and validating the detection models, as well as collecting data in the FFB plantation.

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Conflicts of interest

We, the authors of this manuscript, declare that we have no conflicts of interests.

Generative AI and AI-assisted

We declare that we only used Grammarly for proof reading the paragraphs that we wrote, never used other tools (AI, chat GPT, or other versions) to prepare the research experiments or the manuscript.

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